

# Business Problem Statement

The hospital generates large volumes of patient, appointment, treatment, doctor, and billing data, but lacks a structured analytical approach to evaluate operational performance, patient demand, doctor utilization, treatment costs, and revenue trends. The objective of this project is to analyze healthcare data using SQL to identify high-demand specialties, evaluate doctor appointment and revenue performance, understand patient visit patterns, analyze treatment costs, and track monthly revenue growth. These insights can help hospital management improve resource allocation, optimize doctor utilization, understand patient demand, and identify opportunities for improving financial performance.

## Executive Summary: Healthcare Analytics Project

### Project Overview

This project establishes a database system (HEALTHCARE\_ANALYTICS) to analyze operational performance, clinical utilization, and financial growth across hospital departments. Using advanced SQL techniques—including Common Table Expressions (CTEs), window functions (RANK(), LAG()), multi-table joins, and temporal aggregations—the analysis provides leadership with actionable data insights.

### Key Analytical Areas Covered

- 1. Patient Volume & Demographics**
  - Measured total patient registrations and evaluated the gender-wise patient distribution.
  - Identified high-utilization patients visiting the hospital more than three times to support targeted patient retention and care management strategies.
- 2. Physician Performance & Specialization**
  - Mapped doctor availability across clinical specializations.
  - Evaluated patient appointment volume by doctor and specialization to pinpoint top-performing departments.
  - Isolated physicians whose appointment counts exceed the hospital-wide doctor average.
- 3. Clinical Operations & Treatment Trends**
  - Tracked overall appointment volume and identified seasonal trends using monthly breakdown analysis.
  - Analyzed treatment distribution to determine the most frequently performed procedures alongside their average unit costs.
- 4. Revenue & Financial Growth Management**
  - Calculated cumulative billing revenue generated across hospital operations.
  - Ranked individual physicians based on total revenue contributions derived from treatment costs.
  - Formulated a Month-over-Month (MoM) revenue growth metric using LAG() window functions to measure financial trajectory over time.